

Design Procedure for a 5.8 GHz Rectangular Inset-Fed Microstrip Patch Antenna in CST Studio Suite

Step 1: Create a New Project in CST Studio Suite

1. Launch **CST Studio Suite (Learning Edition)**.
2. Click **New Project**.
3. Under **Application**, select **Microwave & RF / Optical**.
4. Select **Antennas** as the project type.
5. Choose **Planar Antenna** as the antenna template.
6. Select the solver:
 - **Time Domain Solver** (recommended), or
 - **Frequency Domain Solver**.
7. Keep the default unit settings:
 - **Length:** mm
 - **Frequency:** GHz
 - **Time:** ns
8. Set the simulation frequency range:
 - **Minimum Frequency:** 4 GHz
 - **Maximum Frequency:** 8 GHz
9. Enter the **center frequency** as **5.8 GHz**.
10. Click **Finish** to create the project and open the CST modeling environment.

Step 2: Calculate the Initial Antenna Dimensions

Before creating the antenna model in CST Studio Suite, the initial dimensions of the rectangular inset-fed microstrip patch antenna are calculated using the **Microstrip Patch Antenna Dimension Calculator** developed for this project.

The calculator requires the following input parameters:

- Frequency (GHz)
- Dielectric Constant (ϵ_r)
- Substrate Thickness (ST)

RF / MICROWAVE ENGINEERING

500 INSET-FED SYNTHESIS

Microstrip Patch Antenna Dimension Calculator

Analytical synthesis of a rectangular inset-fed microstrip patch antenna from transmission-line theory (Balanis / Hammerstad-Jensen). Enter substrate and frequency parameters to generate starting dimensions for electromagnetic simulation.

TM₁₀₀ MODE

FEED Z₀ = 50Ω (FIXED)

OUTPUT UNITS: mm

PARAMETERS

Frequency GHz

5.8

Dielectric Constant ε_r

2.2

Substrate Thickness (ST) mm

1.6

Copper Thickness (CT) mm

0.035

Defaults reflect standard FR-4 (1.6 mm) with 1 oz copper (0.035 mm). Override per your substrate datasheet.

• Feed impedance fixed internally at 50.00 Ω

DIMENSION TABLE

PARAMETER	VALUE
Ground Length GL	36.08 mm
Ground Width GW	30.05 mm
Substrate Length SL	36.08 mm
Substrate Width SW	30.05 mm
Patch Length PL	16.48 mm
Patch Width PW	20.45 mm
Feed Length FL	10.00 mm
Feed Width FW	4.97 mm
Inset Length IL	4.94 mm
Inset Width IW	4.97 mm

Figure 2.1: *Microstrip Patch Antenna Dimension Calculator Interface.*

After entering the required parameters, click the **Calculate** button. The calculator automatically generates the initial antenna dimensions, including:

- Ground Length (GL)
- Ground Width (GW)
- Substrate Length (SL)
- Substrate Width (SW)
- Patch Length (PL)
- Patch Width (PW)
- Feed Length (FL)
- Feed Width (FW)
- Inset Length (IL)
- Inset Width (IW)

In addition to the antenna dimensions, the calculator provides the **CST Modeling Formula Reference**, which contains the expressions for **X Minimum**, **X Maximum**, **Y Minimum**, **Y Maximum**, **Z Minimum**, and **Z Maximum** for the Ground Plane, Substrate, Patch, Feed Line, and Inset. These expressions can be directly used while creating the geometry using the **Brick** tool in CST Studio Suite, reducing manual calculations and minimizing modeling errors.

The calculated values serve as the initial design dimensions. To achieve the desired resonant frequency and optimum antenna performance, parameters such as **Patch Length (PL)** and **Ground Length (GL)** may be further optimized in CST Studio Suite.

CST MODELING FORMULA REFERENCE		
— GROUND		
$X_{min} = 0$	$Y_{min} = 0$	$Z_{min} = -CT$
$X_{max} = GW$	$Y_{max} = GL$	$Z_{max} = 0$
— SUBSTRATE		
$X_{min} = 0$	$Y_{min} = 0$	$Z_{min} = 0$
$X_{max} = GW$	$Y_{max} = GL$	$Z_{max} = ST$
— PATCH		
$X_{min} = (GW/2) - (PW/2)$	$Y_{min} = FL$	$Z_{min} = ST$
$X_{max} = (GW/2) + (PW/2)$	$Y_{max} = FL + PL$	$Z_{max} = ST + CT$
— FEED LINE		
$X_{min} = (GW/2) - (FW/2)$	$Y_{min} = 0$	$Z_{min} = ST$
$X_{max} = (GW/2) + (FW/2)$	$Y_{max} = FL$	$Z_{max} = ST + CT$
— INSET		
$X_{min} = (GW/2) + (FW/2)$	$Y_{min} = FL$	$Z_{min} = ST$
$X_{max} = (GW/2) + (FW/2) + IW$	$Y_{max} = FL + IL$	$Z_{max} = ST + CT$
Note: These dimensions are analytical starting values and are approximately 90% accurate. For optimum antenna performance, the Patch Length (PL) and Ground Length (GL) should be fine-tuned in CST Studio Suite.		

Figure 2.2: CST Modeling Formula Reference Generated by the Calculator.

Calculator Link: [patch-antenna-calculator](#)

Design Calculation Reference

The dimensions generated by the **Microstrip Patch Antenna Dimension Calculator** are based on standard transmission-line equations for rectangular inset-fed microstrip patch antennas. A separate calculation document has been prepared containing the detailed mathematical derivations, design equations, substitution of values, and optimized antenna parameters used during the design process.

The reference document includes:

- Patch width calculation
- Effective dielectric constant
- Fringing field extension
- Effective and actual patch length calculations
- Ground plane dimension calculations
- Feed width and feed length calculations
- Inset feed optimization
- Final optimized design parameters and CST simulation results for both **5.8 GHz (C-band)** and **1.5 GHz (L-band)** antennas.

[*Design Calculations \(PDF\)*](#)

Step 3: Create the Ground Plane

The ground plane acts as the reference conductor for the microstrip patch antenna and provides a return path for the surface current. It is modeled first because it forms the base of the antenna structure.

Procedure

1. Open **Modeling** → **Shapes** → **Brick**.
2. Create a new brick and rename it **Ground**.
3. Set the material to **Copper (annealed)**.
4. Enter the coordinate values generated by the **Microstrip Patch Antenna Dimension Calculator** in the CST Modeling Formula Reference.
5. Click **OK** to create the ground plane.

The completed ground plane serves as the conducting base on which the substrate, radiating patch, and feed structure will be constructed in the subsequent steps.

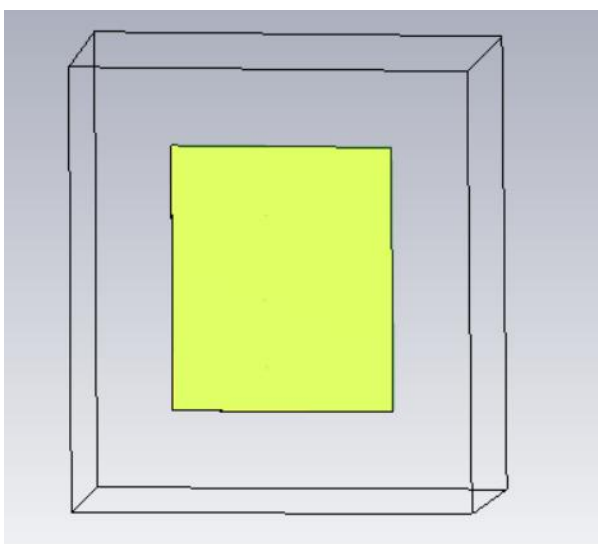


Figure 3.1: Ground Plane Created in CST Studio Suite.

Step 4: Create the Substrate

Similarly, create the substrate using the **Brick** tool. Rename the component as **Substrate** and assign the material as **Rogers RT/Duroid 5880** (or the selected substrate material). Enter the coordinates generated by the **Microstrip Patch Antenna Dimension Calculator** using the CST Modeling Formula Reference.

After entering the coordinates, click **OK** to create the substrate. The substrate should be positioned directly above the ground plane, forming the dielectric layer of the antenna structure.

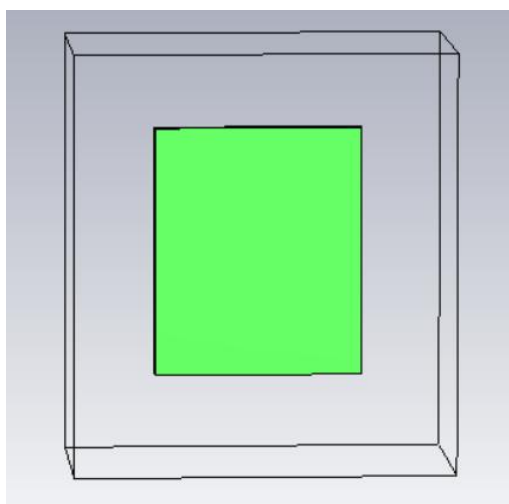


Figure 4.1: Substrate Created in CST Studio Suite.

Step 5: Create the Patch

Similarly, create the radiating patch using the **Brick** tool. Rename the component as **Patch** and assign the material as **Copper (annealed)**. Enter the coordinates generated by the **Microstrip Patch Antenna Dimension Calculator** using the CST Modeling Formula Reference.

After entering the coordinates, click **OK** to create the patch. The patch should be centered on the substrate and positioned above it, serving as the primary radiating element of the antenna.

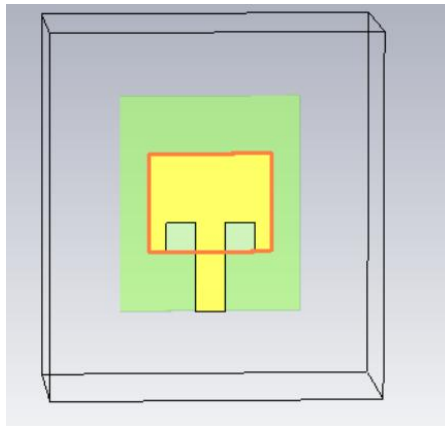


Figure 5.1: Patch Created in CST Studio Suite.

Step 6: Create the Feed Line

Similarly, create the feed line using the **Brick** tool. Rename the component as **Feed Line** and assign the material as **Copper (annealed)**. Enter the coordinates generated by the **Microstrip Patch Antenna Dimension Calculator** using the CST Modeling Formula Reference.

After entering the coordinates, click **OK** to create the feed line. The feed line should extend from the edge of the substrate to the radiating patch and will later be connected to the excitation port.

To ensure electrical continuity between the feed line and the patch, perform a **Boolean ADD** operation. Select both the **Patch** and **Feed Line** components, then navigate to **Modeling** → **Boolean** → **Add** and click **OK**. This merges the two components into a single conducting body, ensuring proper current flow during simulation.

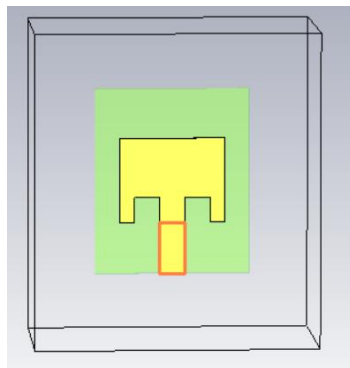


Figure 6.1: Feed Line Created in CST Studio Suite.

Step 7: Create the Inset and Perform Boolean Subtract

Similarly, create the inset using the **Brick** tool. Rename the component as **Inset** and assign the material as **Copper (annealed)**. Enter the coordinates generated by the **Microstrip Patch Antenna Dimension Calculator** using the CST Modeling Formula Reference.

After creating the inset on one side of the feed line, create the second inset by mirroring it about the **X-direction**. Select the inset component and navigate to **Modeling** → **Transform** → **Mirror**. Enable the **Mirror in X-direction** option, check **Copy**, and click **OK**. This creates a symmetric inset on the opposite side of the feed line.

Next, remove the inset regions from the patch by performing a **Boolean Subtract** operation. Select the **Patch-Feed Line** component as the main object and the two inset components as the subtracting objects. Navigate to **Modeling** → **Boolean** → **Subtract** and complete the operation. This forms the final inset-fed microstrip patch antenna geometry.

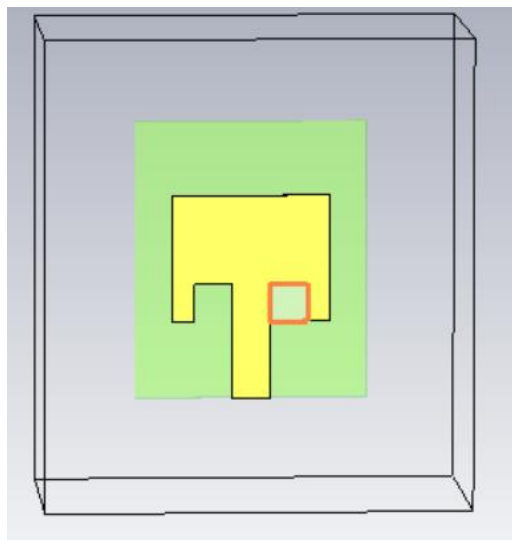


Figure 7.1: Inset Created in CST Studio Suite.

Step 8: Create the Waveguide Port

The waveguide port is used to excite the microstrip feed line and provide the input signal for antenna simulation.

Procedure

1. Zoom in to the **edge of the feed line**.
2. Navigate to **Modeling** → **Pick** → **Pick Face**.
3. Select the metallic face at the end of the feed line by **double-clicking** it. The selected face will be highlighted.
4. Navigate to **Home** → **VBA Macros** → **Solver** → **Ports** → **Calculate Port Extension Coefficient**.
5. In the dialog box, verify the feed line width (**W**), substrate height (**h**), and dielectric constant (**ϵ_r**) are correct.

6. Click **Calculate** to determine the port extension coefficient.
7. Click **Construct Port from Picked Face** to automatically generate the waveguide port.

The waveguide port is now assigned to the feed line and is ready for excitation during simulation.

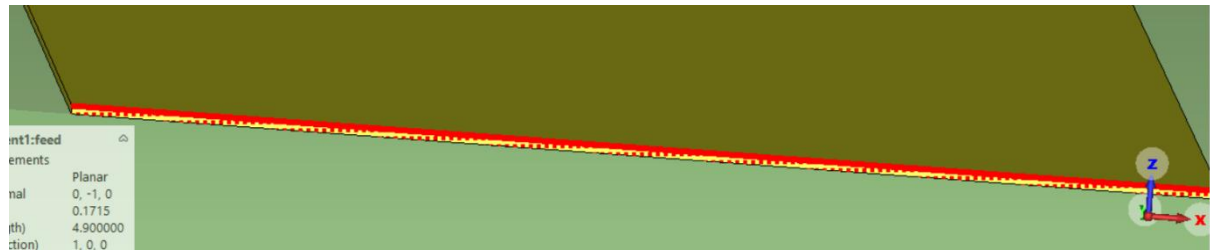


Figure 8.1: Selecting the Feed Line Face.

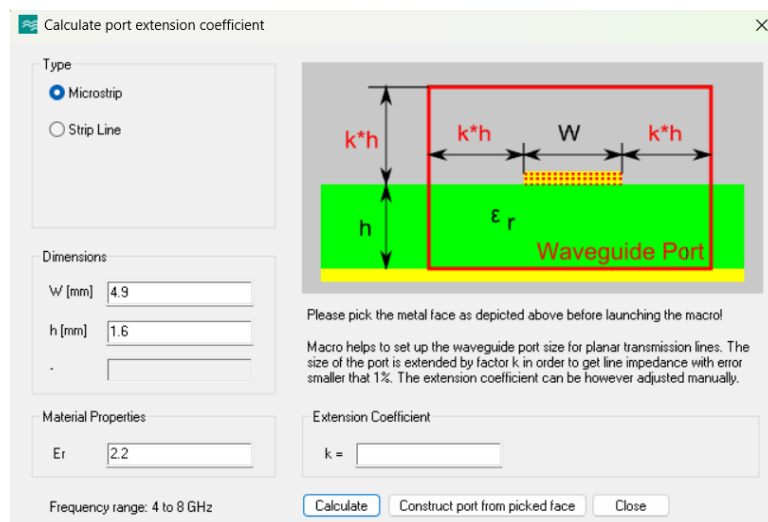


Figure 8.2: Calculate Port Extension Coefficient Dialog.

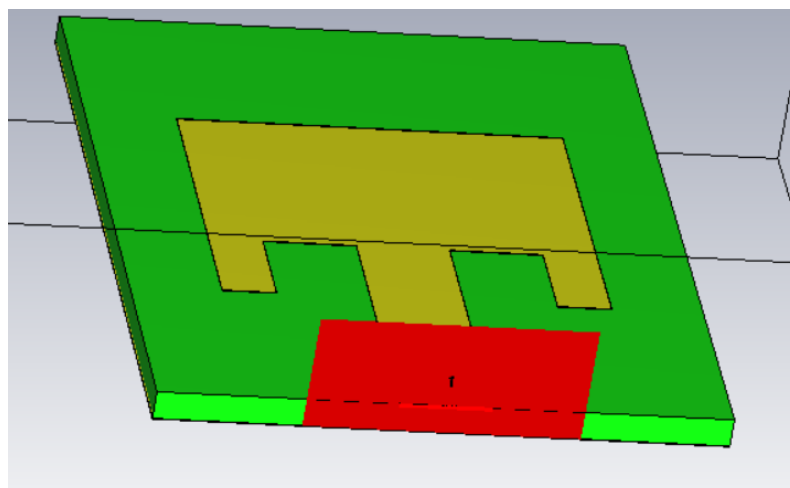


Figure 8.3: Waveguide Port Created at the Feed Line

Step 9: Configure the Farfield Monitor and Run the Simulation

The **Farfield Monitor** is used to calculate the antenna radiation characteristics, such as gain, directivity, and radiation pattern, at the desired operating frequency.

Procedure

1. Navigate to **Simulation** → **Field Monitors**.
2. In the **Monitor** dialog box, select **Farfield/RCS** as the monitor type.
3. Under **Specification**, select **Frequency** and enter the **center frequency** of the antenna (e.g., **5.8 GHz** for the C-band antenna).
4. Click **Apply**, then click **OK** to create the farfield monitor.
5. After configuring the monitor, navigate to **Home** → **Start Simulation** to begin the simulation.

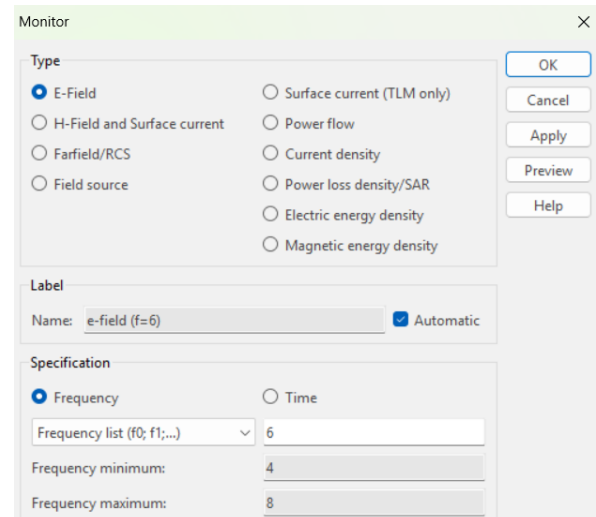


Figure 9.1: Farfield Monitor Configuration.

Once the simulation is complete, CST Studio Suite automatically generates the electromagnetic results, including **S-Parameters (S11)**, **VSWR**, **Gain**, **Directivity**, **Farfield Radiation Pattern**, and **Surface Current Distribution**, which can be analyzed to evaluate the antenna performance.

Step 10: View and Analyze the Simulation Results

After the simulation is completed successfully, the results can be viewed from the **Navigation Tree**.

Procedure

1. Expand the **1D Results** folder to view the antenna's input characteristics.
2. Open the following plots:
 - **S-Parameters (S11)** – to analyze the reflection coefficient and resonant frequency.
 - **VSWR** – to evaluate the impedance matching of the antenna.
3. Expand the **Farfields** folder to analyze the radiation characteristics.
4. Open the following results:
 - **3D Radiation Pattern** – to visualize the antenna radiation pattern.
 - **Gain** – to observe the maximum realized gain.
 - **Directivity** – to evaluate the directional characteristics of the antenna.
 - **Polar Plot** – to examine the radiation pattern in different planes.

These results are used to evaluate the overall performance of the antenna and verify that it meets the desired design specifications.

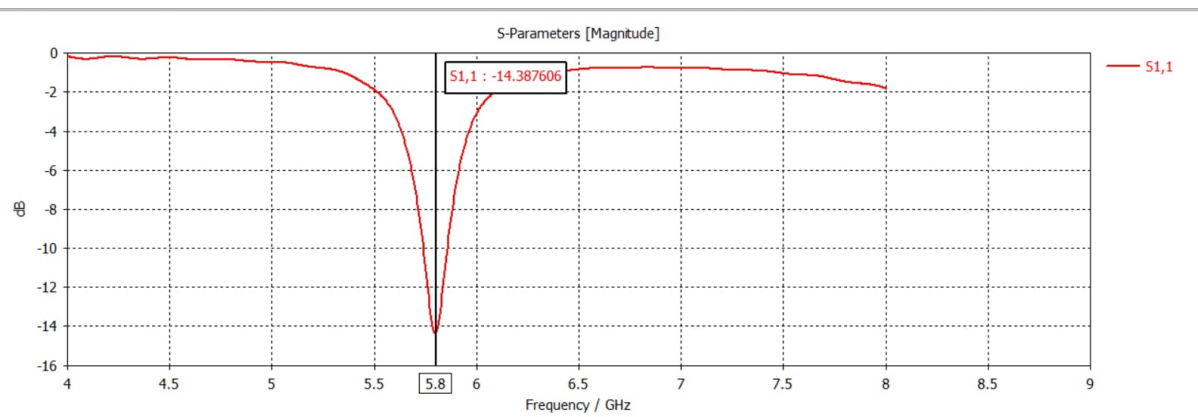


Figure 10.1: S-Parameter (S11) Plot.

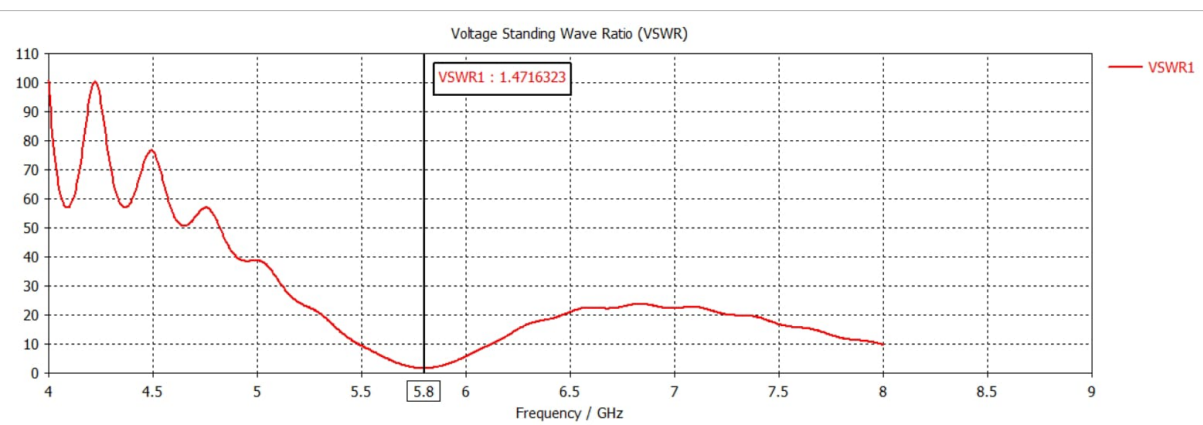


Figure 10.2: VSWR Plot.

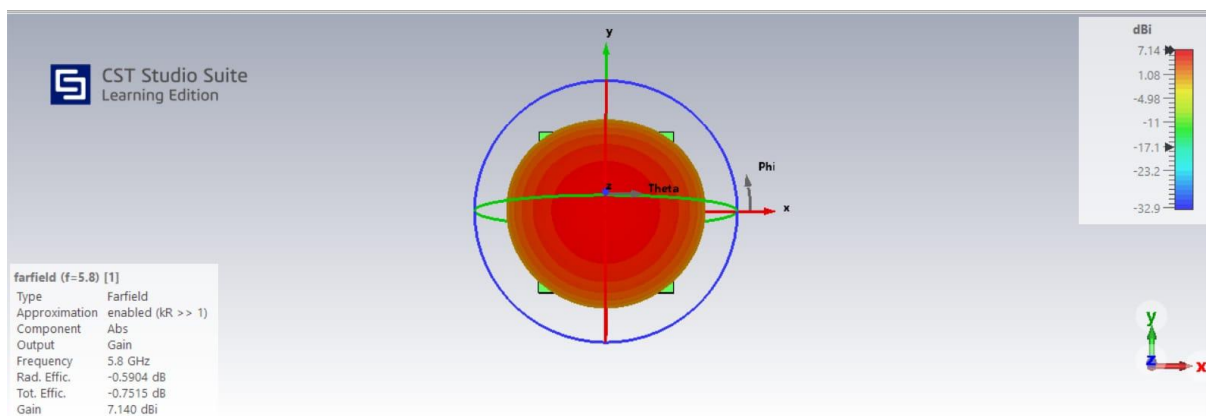


Figure 10.3: Gain Plot.

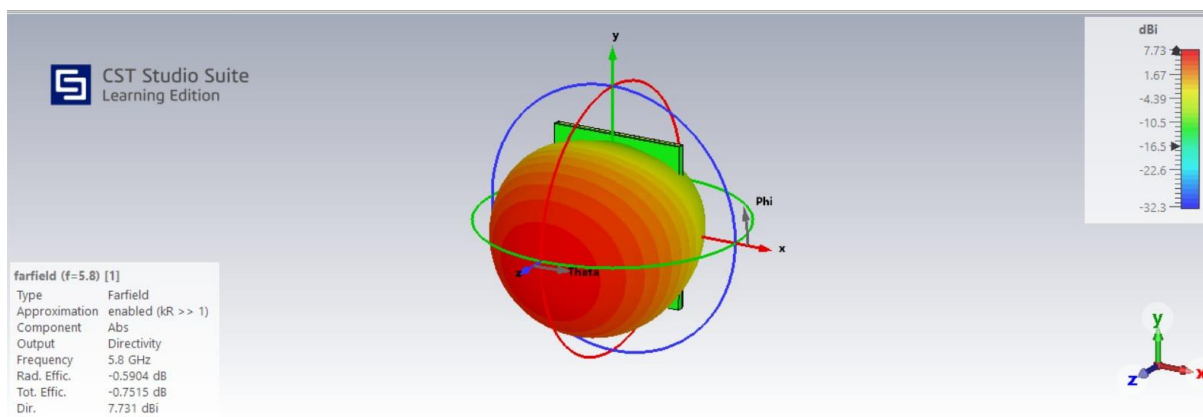


Figure 10.4: Directivity Plot.

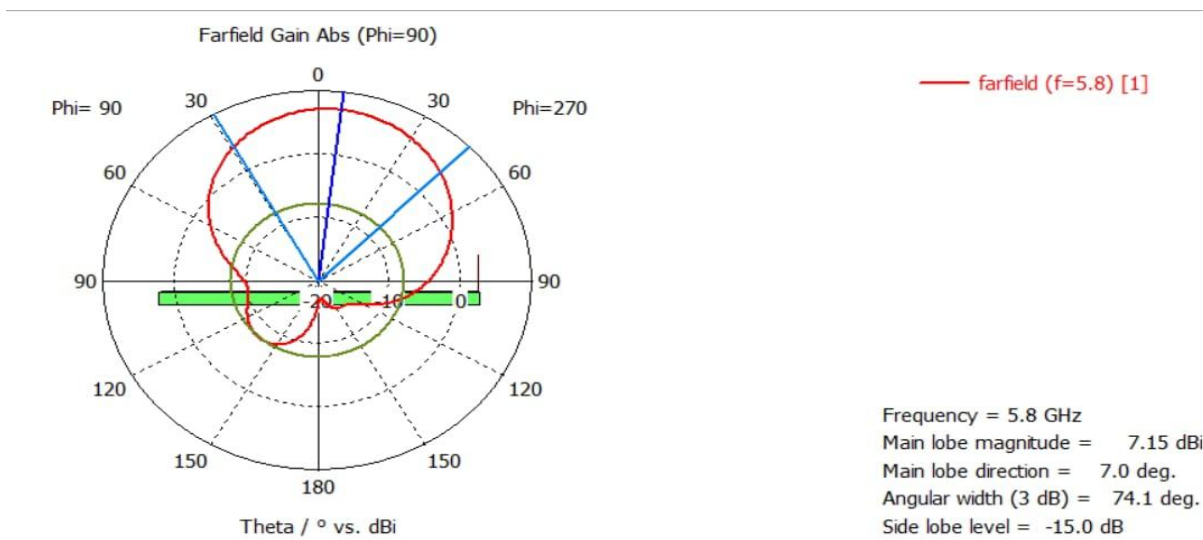


Figure 10.5: Polar Radiation Pattern.

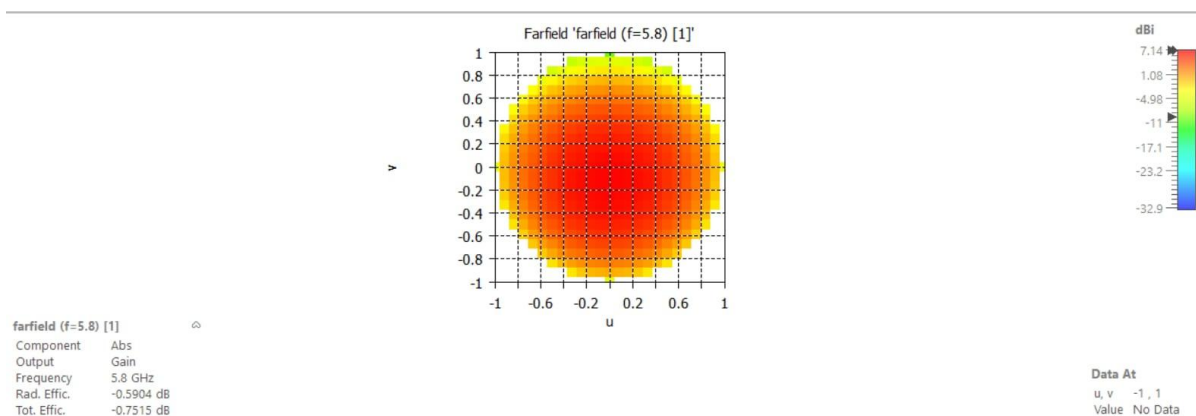


Figure 10.6: 2D Farfield Gain Plot.

Final Antenna Model

The completed 5.8 GHz rectangular inset-fed microstrip patch antenna model is shown below. The geometry was created using the dimensions obtained from the **Microstrip Patch Antenna Dimension Calculator** and modeled in **CST Studio Suite**. The parameter list contains all design variables used to construct the antenna, allowing the model to be easily modified or optimized.

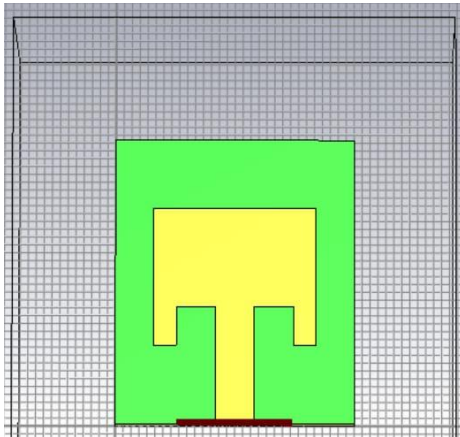


Figure 10.7: Complete Geometry of the 5.8 GHz Rectangular Inset-Fed Microstrip Patch Antenna.

Parameter List			
	Name	Expression	Value
-	wg	= 30.05	30.05
-	lg	= 36.08	36.08
-	ht	= 0.035	0.035
-	hs	= 1.6	1.6
-	wf	= 4.9	4.9
-	lf	= 10	10
-	wp	= 20.4	20.4
-	lp	= 17.4	17.4
-	wc	= 4.9	4.9
-	lc	= 4.94	4.94
< new parameter >			

Figure 10.8: CST Parameter List of the Designed Antenna

Design Procedure for a 1.5 GHz Inset-Fed Rectangular Microstrip Patch Antenna in CST Studio Suite

Step 1: Create a New Project in CST Studio Suite

1. Launch **CST Studio Suite (Learning Edition)**.
2. Click **New Project**.
3. Under **Application**, select **Microwave & RF / Optical**.
4. Select **Antennas** as the project type.
5. Choose **Planar Antenna** as the antenna template.
6. Select the solver:
 - **Time Domain Solver** (recommended), or
 - **Frequency Domain Solver**.
7. Keep the default unit settings:
 - **Length:** mm
 - **Frequency:** GHz
 - **Time:** ns
8. Set the simulation frequency range:
 - **Minimum Frequency:** 1 GHz
 - **Maximum Frequency:** 2 GHz
9. Enter the **center frequency** as **1.5 GHz**.
10. Click **Finish** to create the project and open the CST modeling environment.

Step 2: Calculate the Initial Antenna Dimensions

Before creating the antenna model in CST Studio Suite, the initial dimensions of the rectangular inset-fed microstrip patch antenna are calculated using the **Microstrip Patch Antenna Dimension Calculator** developed for this project.

The calculator requires the following input parameters:

- Frequency (GHz)
- Dielectric Constant (ϵ_r)
- Substrate Thickness (ST)

RF / MICROWAVE ENGINEERING

500 INSET-FED SYNTHESIS

Microstrip Patch Antenna Dimension Calculator

Analytical synthesis of a rectangular inset-fed microstrip patch antenna from transmission-line theory (Balanis / Hammerstad-Jensen). Enter substrate and frequency parameters to generate starting dimensions for electromagnetic simulation.

TM₁₀₀ MODE

FEED Z₀ = 50Ω (FIXED)

OUTPUT UNITS: mm

PARAMETERS

Frequency

1.5

GHz

Dielectric Constant

2.2

ε_r

Substrate Thickness (ST)

1.6

mm

Copper Thickness (CT)

0.035

mm

Defaults reflect standard FR-4 (1.6 mm) with 1 oz copper (0.035 mm). Override per your substrate datasheet.

• Feed impedance fixed internally at 50.00 Ω

DIMENSION TABLE

PARAMETER	VALUE
Ground Length GL	86.30 mm
Ground Width GW	88.66 mm
Substrate Length SL	86.30 mm
Substrate Width SW	88.66 mm
Patch Length PL	66.70 mm
Patch Width PW	79.06 mm
Feed Length FL	10.00 mm
Feed Width FW	4.97 mm
Inset Length IL	20.01 mm
Inset Width IW	4.97 mm

Figure 2.1: *Microstrip Patch Antenna Dimension Calculator Interface.*

After entering the required parameters, click the **Calculate** button. The calculator automatically generates the initial antenna dimensions, including:

- Ground Length (GL)
- Ground Width (GW)
- Substrate Length (SL)
- Substrate Width (SW)
- Patch Length (PL)
- Patch Width (PW)
- Feed Length (FL)
- Feed Width (FW)
- Inset Length (IL)
- Inset Width (IW)

In addition to the antenna dimensions, the calculator provides the **CST Modeling Formula Reference**, which contains the expressions for **X Minimum**, **X Maximum**, **Y Minimum**, **Y Maximum**, **Z Minimum**, and **Z Maximum** for the Ground Plane, Substrate, Patch, Feed Line, and Inset. These expressions can be directly used while creating the geometry using the **Brick** tool in CST Studio Suite, reducing manual calculations and minimizing modeling errors.

The calculated values serve as the initial design dimensions. To achieve the desired resonant frequency and optimum antenna performance, parameters such as **Patch Length (PL)** and **Ground Length (GL)** may be further optimized in CST Studio Suite.

CST MODELING FORMULA REFERENCE		
— GROUND		
$X_{min} = 0$	$Y_{min} = 0$	$Z_{min} = -CT$
$X_{max} = GW$	$Y_{max} = GL$	$Z_{max} = 0$
— SUBSTRATE		
$X_{min} = 0$	$Y_{min} = 0$	$Z_{min} = 0$
$X_{max} = GW$	$Y_{max} = GL$	$Z_{max} = ST$
— PATCH		
$X_{min} = (GW/2) - (PW/2)$	$Y_{min} = FL$	$Z_{min} = ST$
$X_{max} = (GW/2) + (PW/2)$	$Y_{max} = FL + PL$	$Z_{max} = ST + CT$
— FEED LINE		
$X_{min} = (GW/2) - (FW/2)$	$Y_{min} = 0$	$Z_{min} = ST$
$X_{max} = (GW/2) + (FW/2)$	$Y_{max} = FL$	$Z_{max} = ST + CT$
— INSET		
$X_{min} = (GW/2) + (FW/2)$	$Y_{min} = FL$	$Z_{min} = ST$
$X_{max} = (GW/2) + (FW/2) + IW$	$Y_{max} = FL + IL$	$Z_{max} = ST + CT$
Note: These dimensions are analytical starting values and are approximately 90% accurate. For optimum antenna performance, the Patch Length (PL) and Ground Length (GL) should be fine-tuned in CST Studio Suite.		

Figure 2.2: CST Modeling Formula Reference Generated by the Calculator.

Calculator Link: [patch-antenna-calculator](#)

Design Calculation Reference

The dimensions generated by the **Microstrip Patch Antenna Dimension Calculator** are based on standard transmission-line equations for rectangular inset-fed microstrip patch antennas. A separate calculation document has been prepared containing the detailed mathematical derivations, design equations, substitution of values, and optimized antenna parameters used during the design process.

The reference document includes:

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- Feed width and feed length calculations
- Inset feed optimization
- Final optimized design parameters and CST simulation results for both **5.8 GHz (C-band)** and **1.5 GHz (L-band)** antennas.

[*Design Calculations \(PDF\)*](#)

Step 3: Create the Ground Plane

The ground plane acts as the reference conductor for the microstrip patch antenna and provides a return path for the surface current. It is modeled first because it forms the base of the antenna structure.

Procedure

1. Open **Modeling** → **Shapes** → **Brick**.
2. Create a new brick and rename it **Ground**.
3. Set the material to **Copper (annealed)**.
4. Enter the coordinate values generated by the **Microstrip Patch Antenna Dimension Calculator** in the CST Modeling Formula Reference.
5. Click **OK** to create the ground plane.

The completed ground plane serves as the conducting base on which the substrate, radiating patch, and feed structure will be constructed in the subsequent steps.

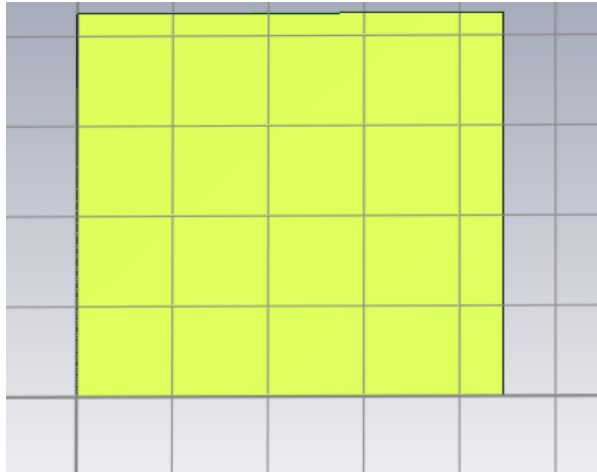


Figure 3.1: Ground Plane Created in CST Studio Suite.

Step 4: Create the Substrate

Similarly, create the substrate using the **Brick** tool. Rename the component as **Substrate** and assign the material as **Rogers RT/Duroid 5880** (or the selected substrate material). Enter the coordinates generated by the **Microstrip Patch Antenna Dimension Calculator** using the CST Modeling Formula Reference.

After entering the coordinates, click **OK** to create the substrate. The substrate should be positioned directly above the ground plane, forming the dielectric layer of the antenna structure.

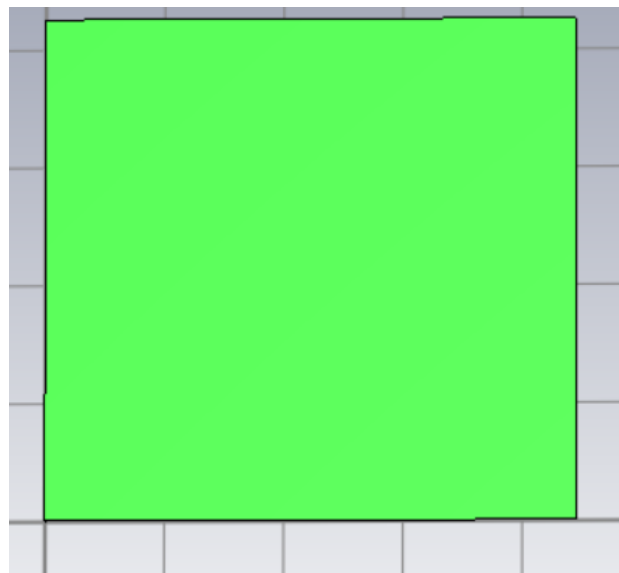


Figure 4.1: Substrate Created in CST Studio Suite.

Step 5: Create the Patch

Similarly, create the radiating patch using the **Brick** tool. Rename the component as **Patch** and assign the material as **Copper (annealed)**. Enter the coordinates generated by the **Microstrip Patch Antenna Dimension Calculator** using the CST Modeling Formula Reference.

After entering the coordinates, click **OK** to create the patch. The patch should be centered on the substrate and positioned above it, serving as the primary radiating element of the antenna.

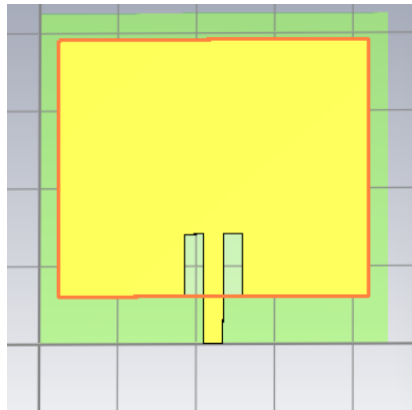


Figure 5.1: Patch Created in CST Studio Suite.

Step 6: Create the Feed Line

Similarly, create the feed line using the **Brick** tool. Rename the component as **Feed Line** and assign the material as **Copper (annealed)**. Enter the coordinates generated by the **Microstrip Patch Antenna Dimension Calculator** using the CST Modeling Formula Reference.

After entering the coordinates, click **OK** to create the feed line. The feed line should extend from the edge of the substrate to the radiating patch and will later be connected to the excitation port.

To ensure electrical continuity between the feed line and the patch, perform a **Boolean ADD** operation. Select both the **Patch** and **Feed Line** components, then navigate to **Modeling** → **Boolean** → **Add** and click **OK**. This merges the two components into a single conducting body, ensuring proper current flow during simulation.

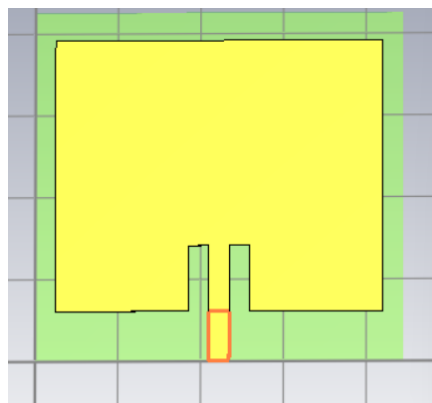


Figure 6.1: Feed Line Created in CST Studio Suite.

Step 7: Create the Inset and Perform Boolean Subtract

Similarly, create the inset using the **Brick** tool. Rename the component as **Inset** and assign the material as **Copper (annealed)**. Enter the coordinates generated by the **Microstrip Patch Antenna Dimension Calculator** using the CST Modeling Formula Reference.

After creating the inset on one side of the feed line, create the second inset by mirroring it about the **X-direction**. Select the inset component and navigate to **Modeling** → **Transform** → **Mirror**. Enable the **Mirror in X-direction** option, check **Copy**, and click **OK**. This creates a symmetric inset on the opposite side of the feed line.

Next, remove the inset regions from the patch by performing a **Boolean Subtract** operation. Select the **Patch-Feed Line** component as the main object and the two inset components as the subtracting objects. Navigate to **Modeling** → **Boolean** → **Subtract** and complete the operation. This forms the final inset-fed microstrip patch antenna geometry.

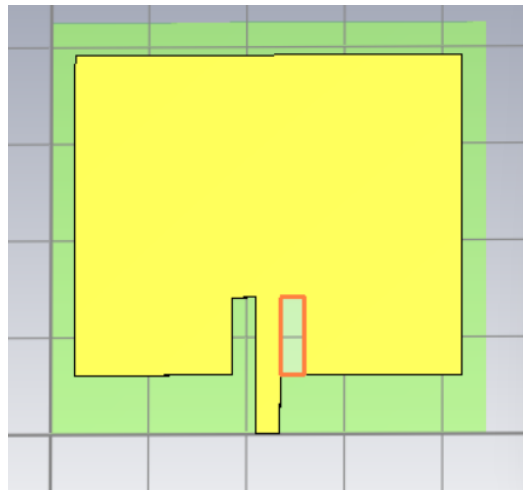


Figure 7.1: Inset Created in CST Studio Suite.

Step 8: Create the Waveguide Port

The waveguide port is used to excite the microstrip feed line and provide the input signal for antenna simulation.

Procedure

1. Zoom in to the **edge of the feed line**.
2. Navigate to **Modeling** → **Pick** → **Pick Face**.
3. Select the metallic face at the end of the feed line by **double-clicking** it. The selected face will be highlighted.
4. Navigate to **Home** → **VBA Macros** → **Solver** → **Ports** → **Calculate Port Extension Coefficient**.
5. In the dialog box, verify the feed line width (**W**), substrate height (**h**), and dielectric constant (**ϵ_r**) are correct.

6. Click **Calculate** to determine the port extension coefficient.
7. Click **Construct Port from Picked Face** to automatically generate the waveguide port.

The waveguide port is now assigned to the feed line and is ready for excitation during simulation.

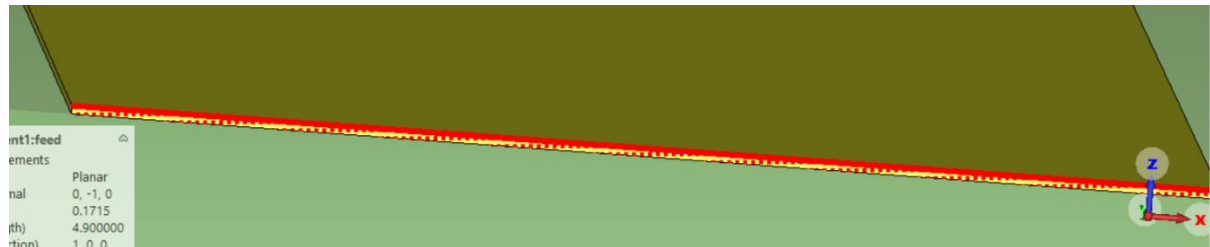


Figure 8.1: Selecting the Feed Line Face.

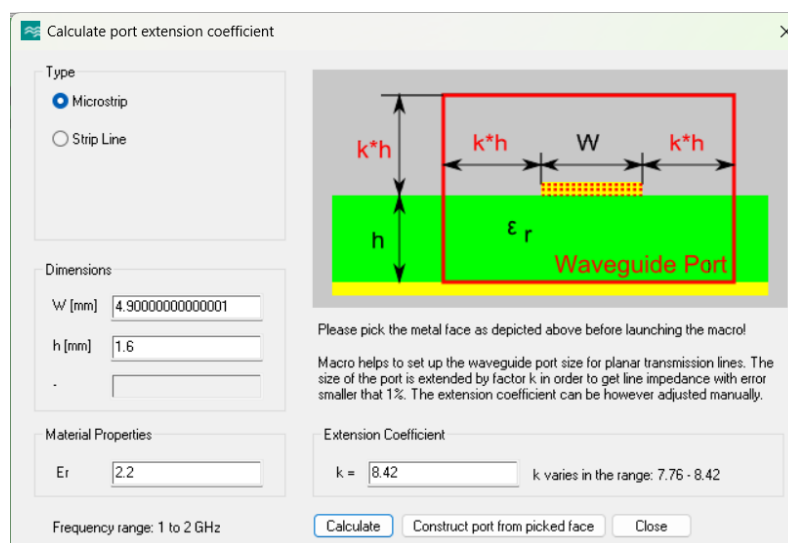


Figure 8.2: Calculate Port Extension Coefficient Dialog.

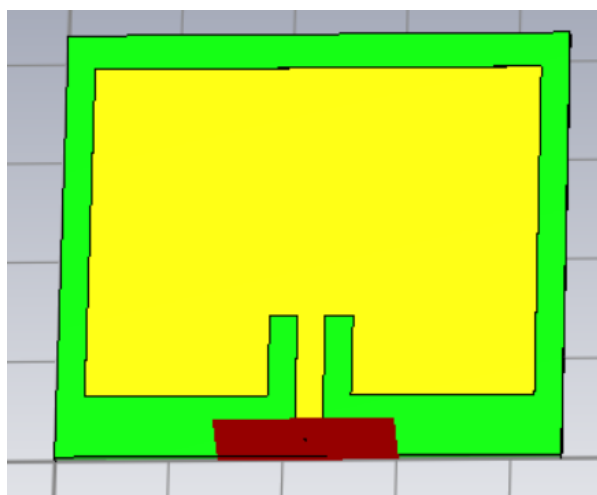


Figure 8.3: Waveguide Port Created at the Feed Line

Step 9: Configure the Farfield Monitor and Run the Simulation

The **Farfield Monitor** is used to calculate the antenna radiation characteristics, such as gain, directivity, and radiation pattern, at the desired operating frequency.

Procedure

1. Navigate to **Simulation** → **Field Monitors**.
2. In the **Monitor** dialog box, select **Farfield/RCS** as the monitor type.
3. Under **Specification**, select **Frequency** and enter the **center frequency** of the antenna (e.g., **1.5 GHz** for the L-band antenna).
4. Click **Apply**, then click **OK** to create the farfield monitor.
5. After configuring the monitor, navigate to **Home** → **Start Simulation** to begin the simulation.

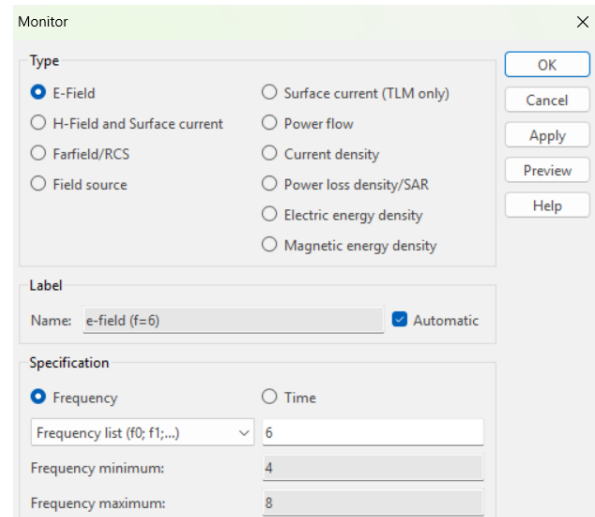


Figure 9.1: Farfield Monitor Configuration.

Once the simulation is complete, CST Studio Suite automatically generates the electromagnetic results, including **S-Parameters (S11)**, **VSWR**, **Gain**, **Directivity**, **Farfield Radiation Pattern**, and **Surface Current Distribution**, which can be analyzed to evaluate the antenna performance.

Step 10: View and Analyze the Simulation Results

After the simulation is completed successfully, the results can be viewed from the **Navigation Tree**.

Procedure

1. Expand the **1D Results** folder to view the antenna's input characteristics.
2. Open the following plots:
 - **S-Parameters (S11)** – to analyze the reflection coefficient and resonant frequency.
 - **VSWR** – to evaluate the impedance matching of the antenna.
3. Expand the **Farfields** folder to analyze the radiation characteristics.
4. Open the following results:
 - **3D Radiation Pattern** – to visualize the antenna radiation pattern.
 - **Gain** – to observe the maximum realized gain.
 - **Directivity** – to evaluate the directional characteristics of the antenna.
 - **Polar Plot** – to examine the radiation pattern in different planes.

These results are used to evaluate the overall performance of the antenna and verify that it meets the desired design specifications.

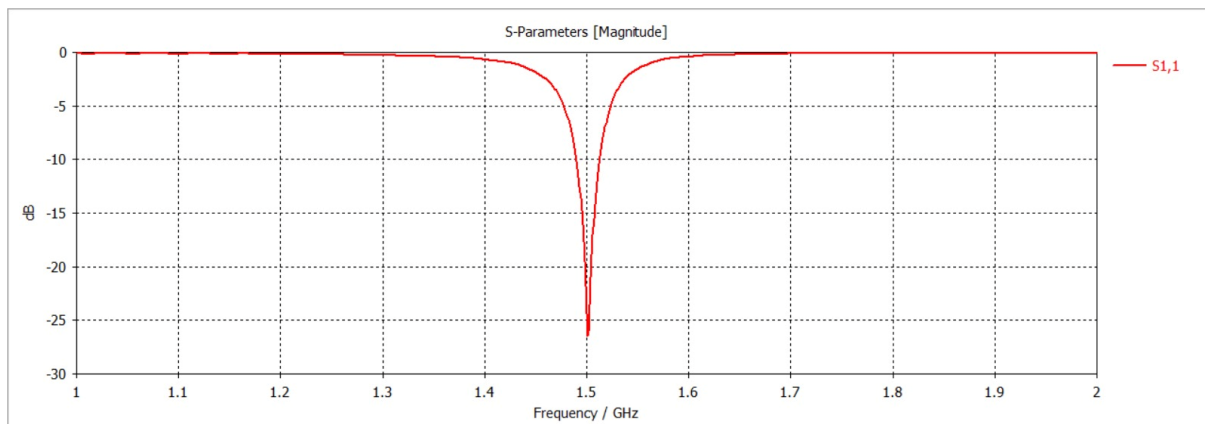


Figure 10.1: S-Parameter (S_{11}) Plot.

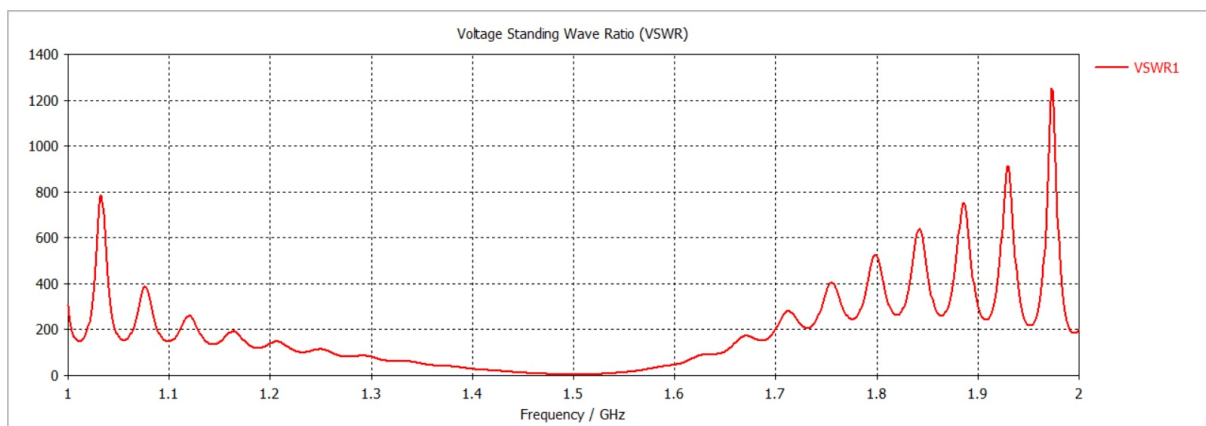


Figure 10.2: VSWR Plot.

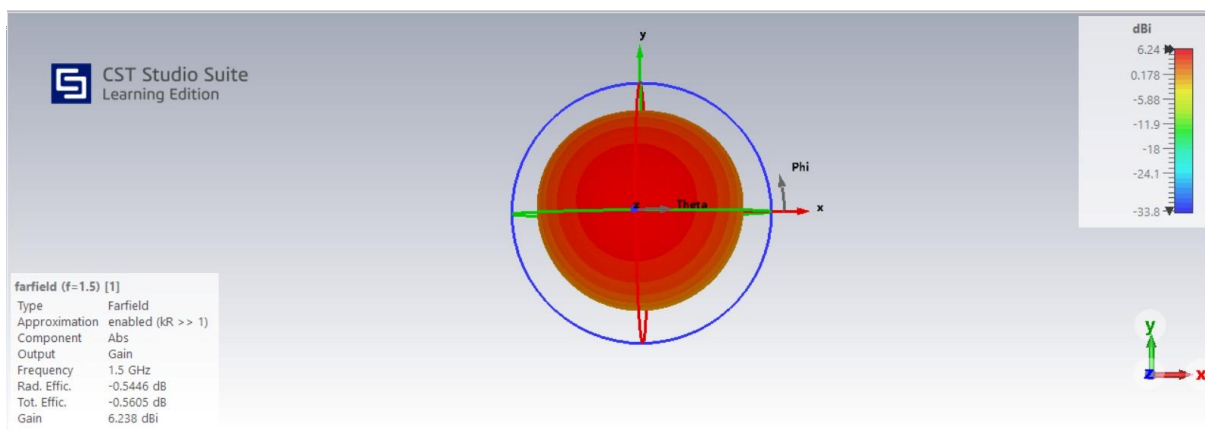


Figure 10.3: Gain Plot.

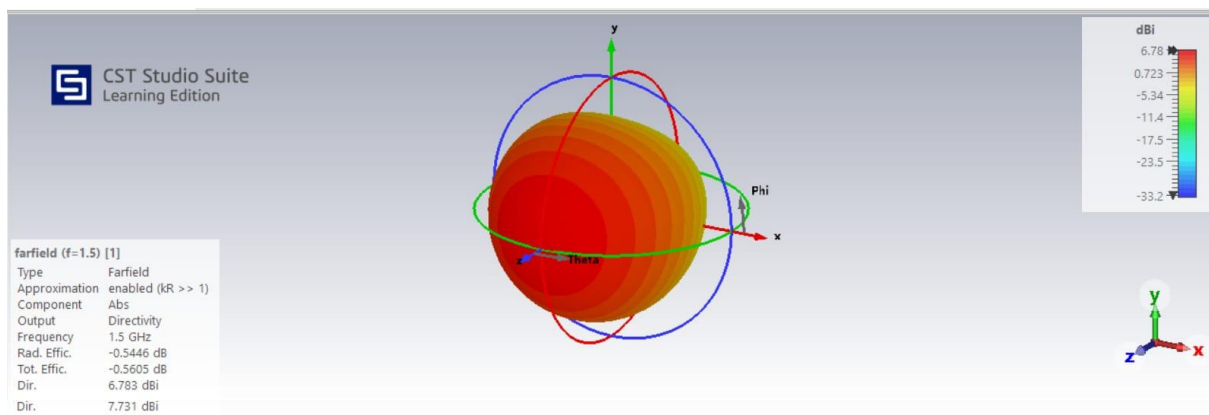


Figure 10.4: Directivity Plot.

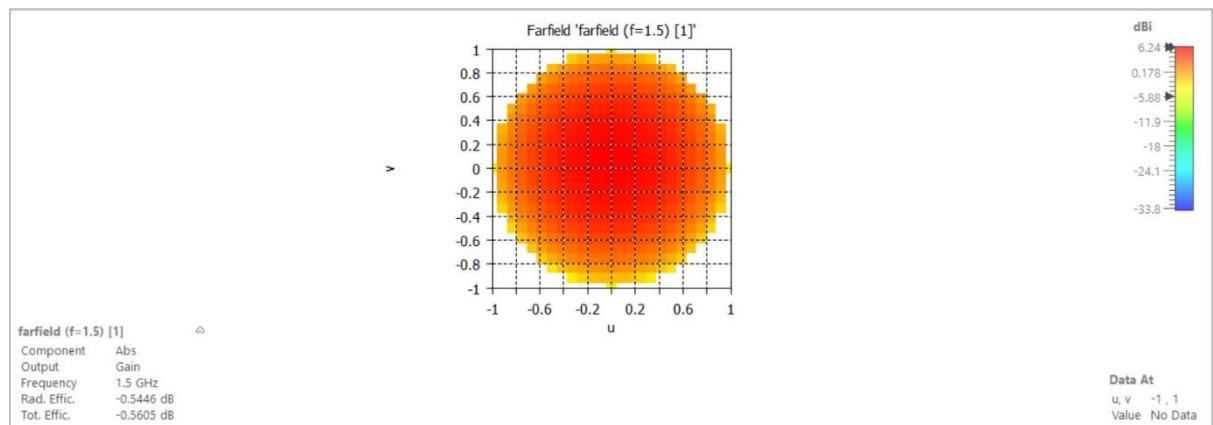


Figure 10.6: 2D Farfield Gain Plot.

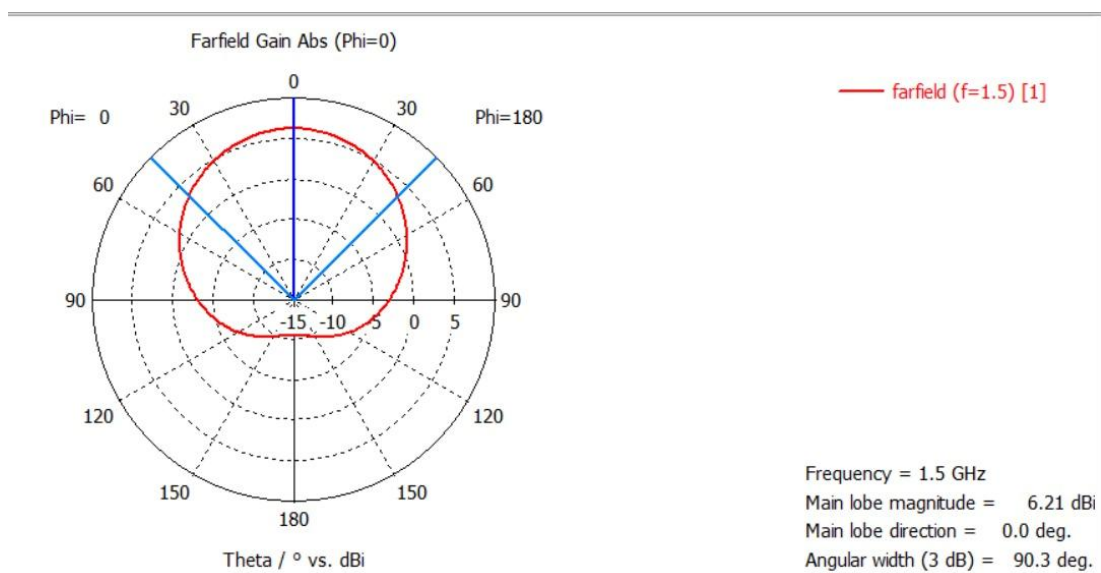


Figure 10.5: Polar Radiation Pattern.

Final Antenna Model

The completed 1.5 GHz rectangular inset-fed microstrip patch antenna model is shown below. The geometry was created using the dimensions obtained from the **Microstrip Patch Antenna Dimension Calculator** and modeled in **CST Studio Suite**. The parameter list contains all design variables used to construct the antenna, allowing the model to be easily modified or optimized.

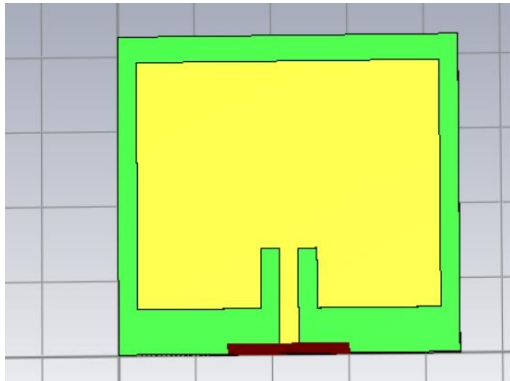


Figure 10.7: Complete Geometry of the 1.5 GHz Rectangular Inset-Fed Microstrip Patch Antenna.

Name	Expression	Value	Description
lf	= 12	12	feed length
wf	= 4.9	4.9	feed width
ht	= 0.035	0.035	ground height
lg	= 85	85	ground length
wg	= 89	89	ground width
lc	= 16	16	inset length
wc	= 4.9	4.9	inset width
lp	= 66.1	66.1	patch length
wp	= 79.1	79.1	patch width
hs	= 1.6	1.6	

Figure 10.8: CST Parameter List of the Designed Antenna